

Simple Linear Regression

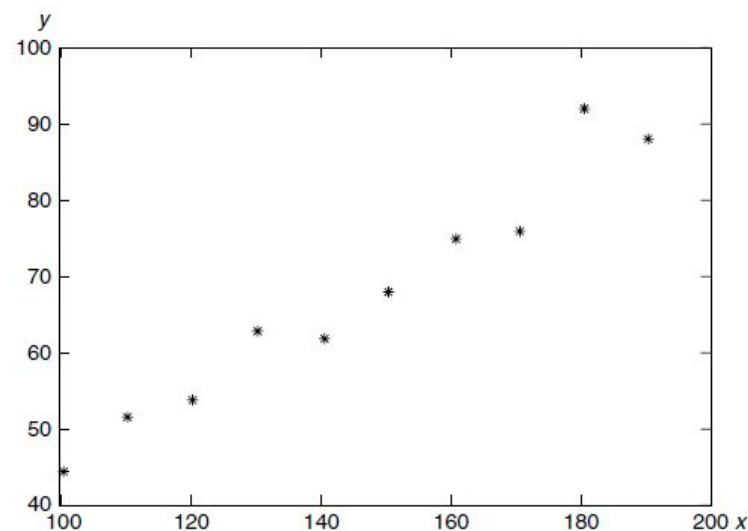
Correlation

- Under the circumstances where one random variable is related to other, there exist a degree of relationship between the variables with reference to the measured observations.
- In statistics such a relationship is known as correlation
- If there are only two random variables , then the correlation between them is called simple correlation

Scatter diagram

EXAMPLE 9.1a Consider the following 10 data pairs (x_i, y_i) , $i = 1, \dots, 10$, relating y , the percent yield of a laboratory experiment, to x , the temperature at which the experiment was run.

i	x_i	y_i	i	x_i	y_i
1	100	45	6	150	68
2	110	52	7	160	75
3	120	54	8	170	76
4	130	63	9	180	92
5	140	62	10	190	88

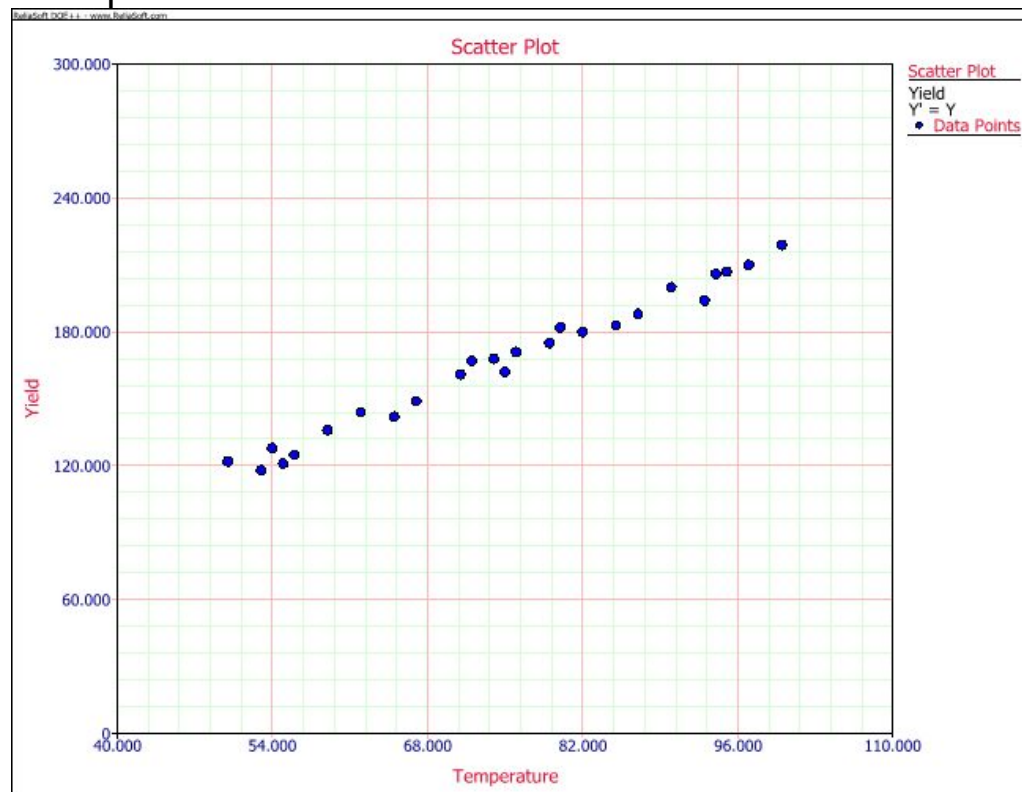


Regression

- If two random variables are found to be correlated , then we can expect that one of the variables (say input variable) influences the other variable (say output variable)
- Such a relationship may be linear or nonlinear
- The influencing variable is called *independent variable*
- The variable that is influenced is called *dependant variable*
- In statistics the independent variable is called as *regressor*
- The dependent variable is also referred to as the *response*

Simple Regression

- A linear regression model attempts to explain the relationship between two or more variables using a straight line
- In the scatter plot yield y_i , is plotted for different temperature values x_i .



Simple Regression

- It is clear that no line can be found to pass through all points of the plot.
- Thus no functional relation exists between the two variables X and Y.
- However, the scatter plot does give an indication that a straight line may exist such that all the points on the plot are scattered randomly around this line.
- A statistical relation is said to exist in this case.
- The statistical relation between X and Y may be expressed as follows:

$$Y = \beta_0 + \beta_1 x + e$$

- Quantities β_0 and β_1 are called *regression coefficients*

Slope and intercept of a line

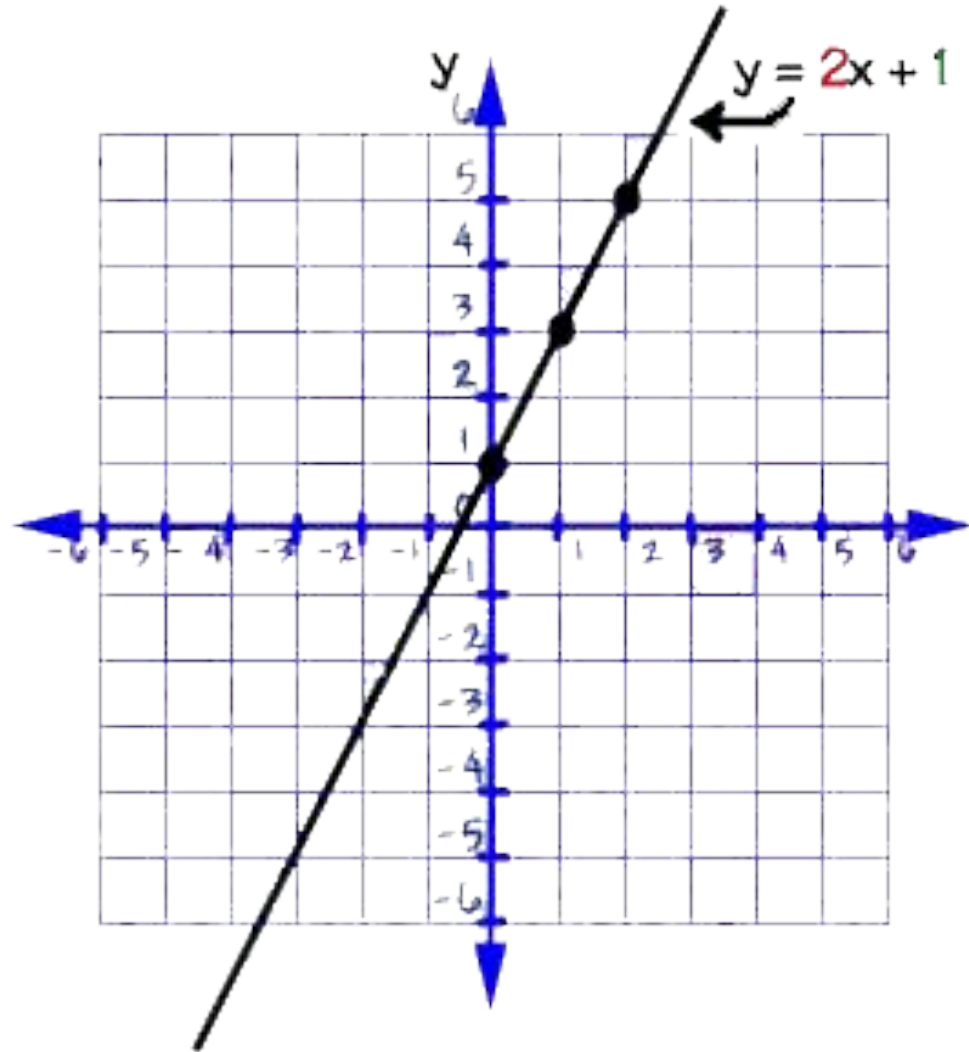
- Every straight line can be represented by an equation:
 $y = mx + b$.
- The coordinates of every point on the line will solve the equation if you substitute them in the equation for x and y .
- The slope m of this line - its steepness, or slant - can be calculated like this:

$$m = \frac{\text{change in y-value}}{\text{change in x-value}}$$

- The equation of any straight line, called a linear equation, can be written as: $y = mx + b$,
- where m is the slope of the line and b is the y-intercept.
- The y-intercept of this line is the value of y at the point where the line crosses the y axis.

Slope and intercept of a line

- $Y=mx+b$
- $m=2$
- $b=1$



Simple Regression

- The equation $Y = \beta_0 + \beta_1 x + e$ is the *linear regression model* that can be used to explain the relation between X and Y that is seen on the scatter plot.
- In this model, the mean value of Y (abbreviated as $E(Y)$) is assumed to follow the linear relation:

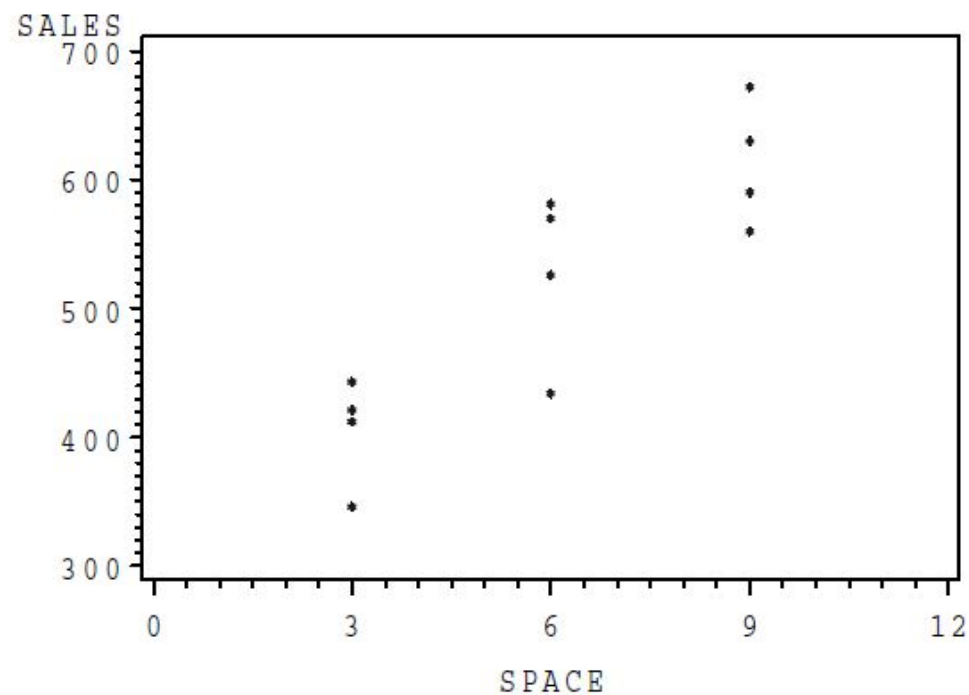
$$E(Y) = \beta_0 + \beta_1 x$$
$$Y = E(Y) + e$$

Least Squares Estimation of β_0 and β_1

- A marketer is interested in the relation between the width of the shelf space for her brand of coffee (x) and weekly sales (y) of the product in a suburban supermarket (assume the height is always at eye level). She believes that in the range of 3 to 9 feet, the **mean weekly sales** will be linearly related to the width of the shelf space. Suppose the marketer conducted the experiment over a twelve week period (4 weeks with 3' of shelf space, 4 weeks with 6', and 4 weeks with 9'), and observed the sample data

Shelf Space	Weekly Sales	Shelf Space	Weekly Sales
x	y	x	y
6	526	6	434
3	421	3	443
6	581	9	590
9	630	6	570
3	412	3	346
9	560	9	672

Table 1: Coffee sales data for $n = 12$ weeks



Least Squares Estimation of β_0 and β_1

- We now have to use sample data to compute estimates of the parameters β_0 and β_1
- If we define the **fitted equation** to be an equation:

$$\hat{y} = \beta_0 + \beta_1 x$$

- We can choose the estimates β_0 and β_1 to be the values that minimize the distances of the data points to the fitted line.

- For each observed response y_i , with a corresponding predictor variable x_i , we obtain a **fitted value** $\hat{y}_i = b_0 + b_1x_i$
- We would like to minimize the sum of the squared distances of each observed response to its fitted value
- That is, we want to minimize the **error sum of squares**, SSE, where:

$$SSE = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \sum_{i=1}^n (y_i - (b_0 + b_1x_i))^2$$

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

Observed Value
of Y for X_i

Predicted Value
of Y for X_i

Intercept = β_0

ε_i

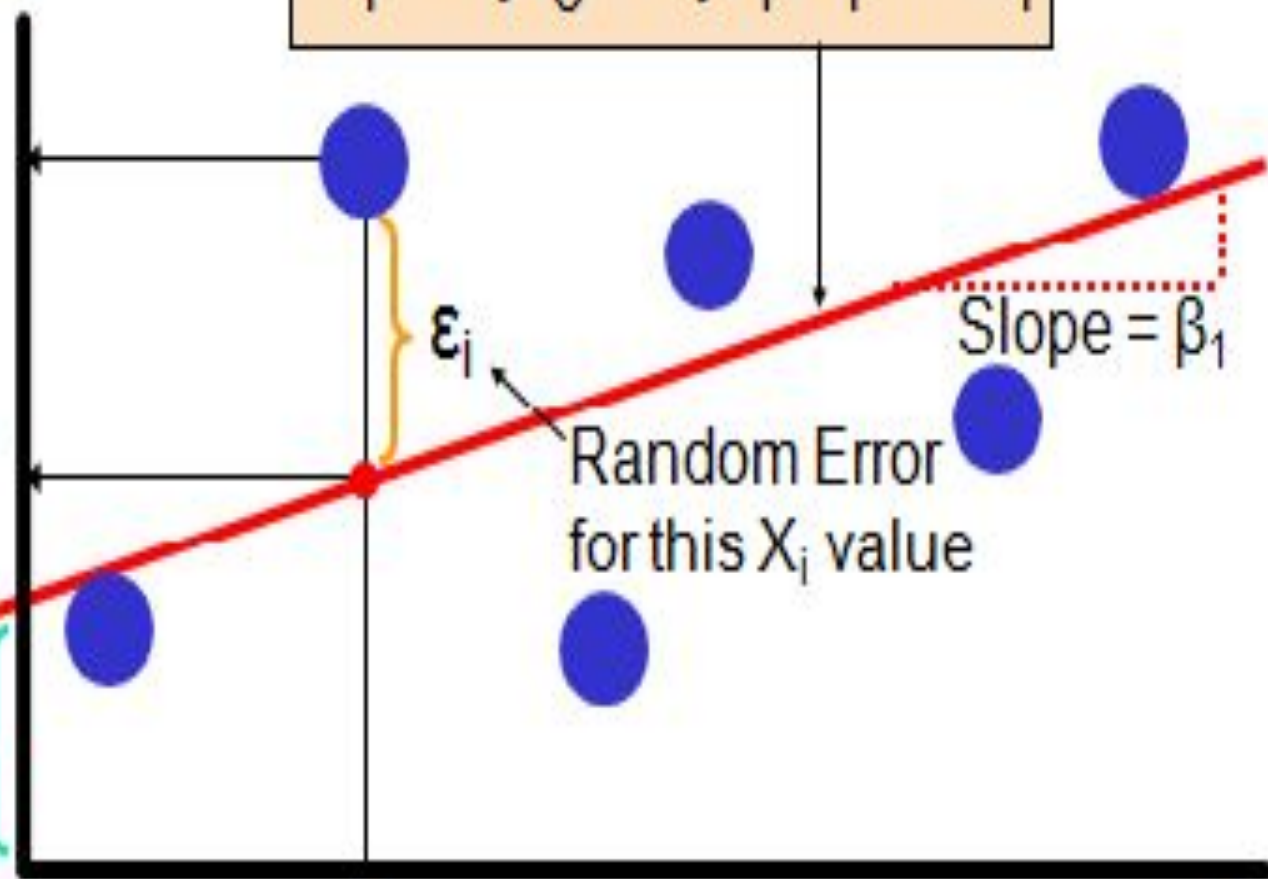
Random Error
for this X_i value

Slope = β_1

X_i

X

Y



- If A is the estimator of β_0 and B of β_1 , then the estimator of the response corresponding to the input variable x_i would be $A + Bx_i$
- **Sum of squares**, SSE will be

$$SS = \sum_{i=1}^n (Y_i - A - Bx_i)^2$$

- To determine these estimators, we differentiate SS first with respect to A and then to B

- Differentiating SS first with respect to A and then to B

$$\frac{\partial SS}{\partial A} = -2 \sum_{i=1}^n (Y_i - A - Bx_i)$$

$$\frac{\partial SS}{\partial B} = -2 \sum_{i=1}^n x_i (Y_i - A - Bx_i)$$

- Setting these partial derivatives equal to zero yields the equations for A and B and are

$$\sum_{i=1}^n Y_i = nA + B \sum_{i=1}^n x_i$$

values A
Equation 1
conditions.

$$\sum_{i=1}^n x_i Y_i = A \sum_{i=1}^n x_i + B \sum_{i=1}^n x_i^2$$

Eqn 2

- Let $\bar{Y} = \sum_i Y_i/n, \quad \bar{x} = \sum_i x_i/n$
- Then we can write the first normal equation as
- Substituting this $A = \bar{Y} - B\bar{x}$ into the second normal equation yields

$$\sum_i x_i Y_i = (\bar{Y} - B\bar{x})n\bar{x} + B \sum_i x_i^2$$

or

$$B \left(\sum_i x_i^2 - n\bar{x}^2 \right) = \sum_i x_i Y_i - n\bar{x}\bar{Y}$$

or

$$B = \frac{\sum_i x_i Y_i - n\bar{x}\bar{Y}}{\sum_i x_i^2 - n\bar{x}^2}$$

- Hence the value of the regression coefficients can be expressed as

$$b_1 = \frac{SS_{xy}}{SS_{xx}},$$

$$b_0 = \bar{y} - \hat{\beta}_1 \bar{x} = \frac{\sum_{i=1}^n y_i}{n} - b_1 \frac{\sum_{i=1}^n x_i}{n}.$$

- where

- $SS_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2 = \sum_{i=1}^n x_i^2 - \frac{(\sum_{i=1}^n x_i)^2}{n}$

- $SS_{xy} = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) = \sum_{i=1}^n x_i y_i - \frac{(\sum_{i=1}^n x_i)(\sum_{i=1}^n y_i)}{n}$

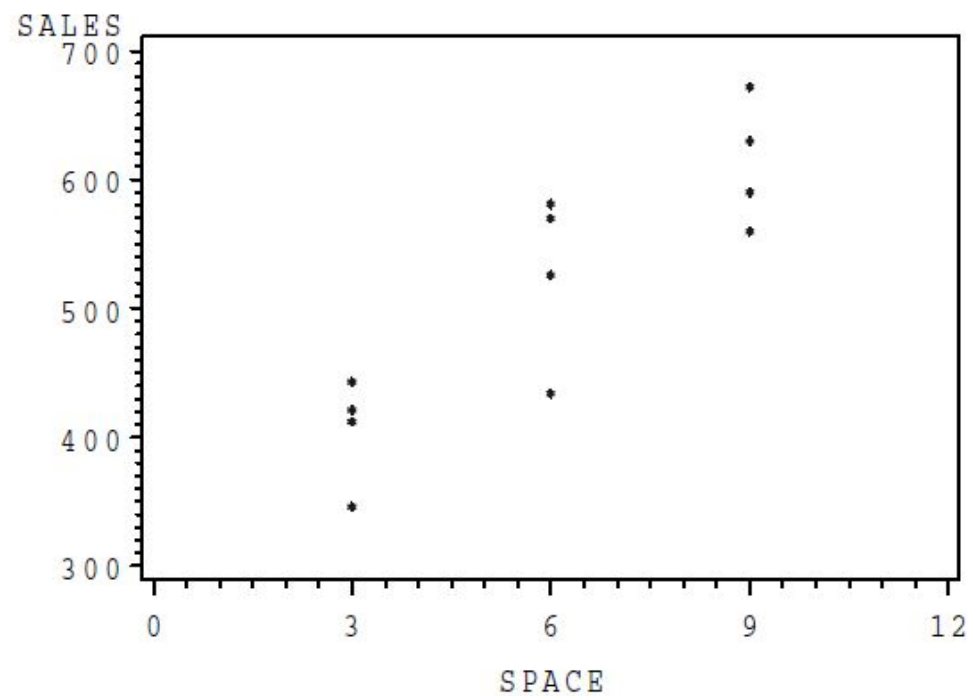
- $SS_{yy} = \sum_{i=1}^n (y_i - \bar{y})^2 = \sum_{i=1}^n y_i^2 - \frac{(\sum_{i=1}^n y_i)^2}{n}$

Problem

- A marketer is interested in the relation between the width of the shelf space for her brand of coffee (x) and weekly sales (y) of the product in a suburban supermarket (assume the height is always at eye level). She believes that in the range of 3 to 9 feet, the **mean weekly sales** will be linearly related to the width of the shelf space. Suppose the marketer conducted the experiment over a twelve week period (4 weeks with 3' of shelf space, 4 weeks with 6', and 4 weeks with 9'), and observed the sample data

Shelf Space	Weekly Sales	Shelf Space	Weekly Sales
x	y	x	y
6	526	6	434
3	421	3	443
6	581	9	590
9	630	6	570
3	412	3	346
9	560	9	672

Table 1: Coffee sales data for $n = 12$ weeks



Week	Space (x)	Sales (y)	x^2	xy	y^2
1	6	526	36	3156	276676
2	3	421	9	1263	177241
3	6	581	36	3486	337561
4	9	630	81	5670	396900
5	3	412	9	1236	169744
6	9	560	81	5040	313600
7	6	434	36	2604	188356
8	3	443	9	1329	196249
9	9	590	81	5310	348100
10	6	570	36	3420	324900
11	3	346	9	1038	119716
12	9	672	81	6048	451584
$\sum x = 72$		$\sum y = 6185$	$\sum x^2 = 504$	$\sum xy = 39600$	$\sum y^2 = 3300627$

$$SS_{xx} = \sum (x - \bar{x})^2 = \sum x^2 - \frac{(\sum x)^2}{n} = 504 - \frac{(72)^2}{12} = 72$$

$$SS_{xy} = \sum (x - \bar{x})(y - \bar{y}) = \sum xy - \frac{(\sum x)(\sum y)}{n} = 39600 - \frac{(72)(6185)}{12} = 2490$$

$$SS_{yy} = \sum (y - \bar{y})^2 = \sum y^2 - \frac{(\sum y)^2}{n} = 3300627 - \frac{(6185)^2}{12} = 112772.9$$

- We obtain the least squares estimate of the true linear regression relation $Y = \beta_0 + \beta_1 x$

$$b_1 = \frac{SS_{xy}}{SS_{xx}} = \frac{2490}{72} = 34.5833$$

$$b_0 = \frac{\sum y}{n} - b_1 \frac{\sum x}{n} = \frac{6185}{12} - 34.5833 \left(\frac{72}{12} \right) = 515.4167 - 207.5000 = 307.967.$$

$$\hat{y} = b_0 + b_1 x = 307.967 + 34.583x$$

- So the fitted equation, estimating the mean weekly sales when the product has x feet of shelf space is $\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 x = 307.967 + 34.5833x$.

THE COEFFICIENT OF DETERMINATION

THE COEFFICIENT OF DETERMINATION

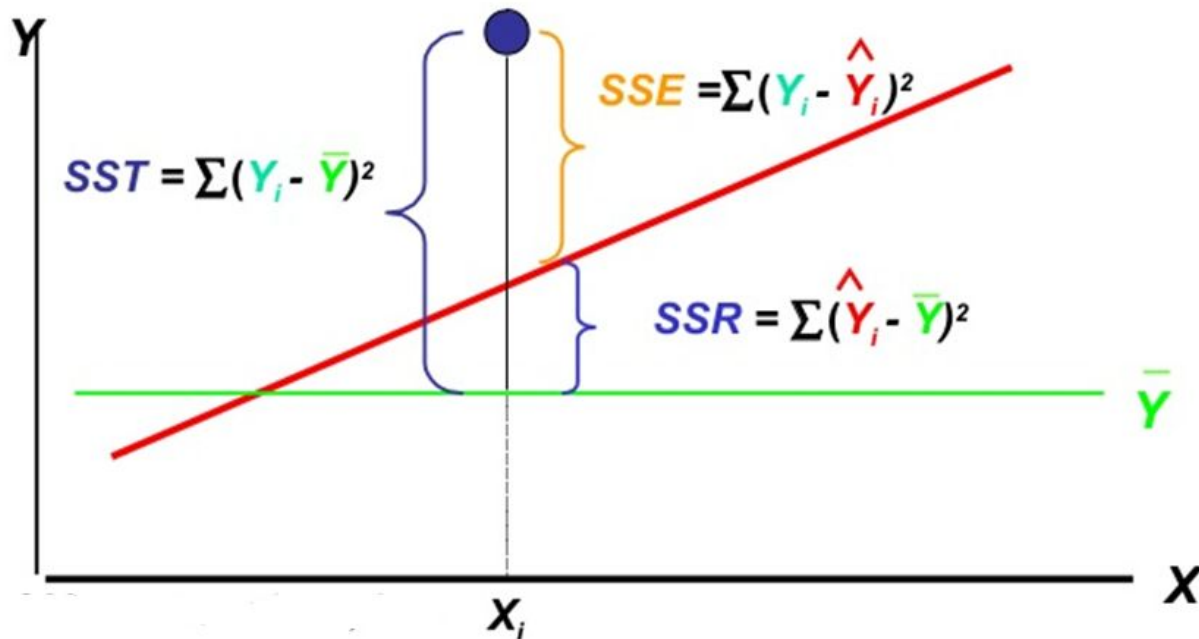
- Suppose we wanted to measure the amount of variation in the set of response values $Y_1, \dots, Y_n$ corresponding to the set of input values $x_1, \dots, x_n$
- A standard measure in statistics of the amount of variation in a set of values $Y_1, \dots, Y_n$ is given by the quantity
- If all the Y_i are equal — and thus are all equal to Y — then S_{YY} would equal 0.

- Let us consider now the question as to how much of the variation
 - in the values of the response variables is due to the different input values, and
 - how much is due to the inherent variance of the responses even when the input values are taken into account
- To answer the query, consider the quantity

$$SS_R = \sum_{i=1}^n (Y_i - A - Bx_i)^2$$

- $\sum (y_i - \bar{y})^2 = TSS =$ Total Sum Squared = SS_{yy}
- $\sum (\hat{y}_i - \bar{\hat{y}})^2 = ESS =$ Explained Sum Squared
- $\sum (e_i^2) = SSR =$ Sum of Squared Residuals

The Sum of Squares



- The proportion of the variation in the response variables that is explained by the different input values is given by

$$\begin{aligned} R^2 &= \frac{S_{YY} - SS_R}{S_{YY}} \\ &= 1 - \frac{SS_R}{S_{YY}} \end{aligned}$$

- R^2 is called the *coefficient of determination*
- The coefficient of determination R^2 will have a value between 0 and 1.
- A value of R^2 near 1 indicates that most of the variation of the response data is explained by the different input values whereas a value of R^2 near 0 indicates that little of the variation is explained by the different input values
- The value of R^2 is often used as an indicator of how well the regression model fits the data, with a value near 1 indicating a good fit, and one near 0 indicating a poor fit

Multiple Linear Regression

Multiple Linear Regression

- In the majority of applications, the **response** of an experiment **can be predicted** more adequately **not on the basis of a single independent** input variable **but on a collection of such variables**.
- Indeed, a typical situation is one in which there are a set of, say, k input variables and the response Y is related to them by the relation

$$Y = \beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k + e$$

Multiple Linear Regression

EXAMPLE 9.10a The data in Table 9.4 relate the suicide rate to the population size and the divorce rate at eight different locations.

TABLE 9.4

Location	Population in Thousands	Divorce Rate per 100,000	Suicide Rate per 100,000
Akron, OH	679	30.4	11.6
Anaheim, CA	1,420	34.1	16.1
Buffalo, NY	1,349	17.2	9.3
Austin, TX	296	26.8	9.1
Chicago, IL	6,975	29.1	8.4
Columbia, SC	323	18.7	7.7
Detroit, MI	4,200	32.6	11.3
Gary, IN	633	32.5	8.4

$$Y = \beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k + e$$

Multiple Linear Regression

$$Y = \beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k + e$$

- where x_j , $j = 1, \dots, k$ is the level of the j th input variable and e is an error.
- The parameters $\beta_0, \beta_1, \dots, \beta_k$ are assumed to be unknown and must be estimated from the data, which we shall suppose will consist of the values of $Y_1, \dots, Y_n$ where Y_i is the response level corresponding to the k input levels $x_{i1}, \dots, x_{i2}, \dots, x_{ik}$.
- That is, the Y_i are related to these input levels through

$$\hat{y}_i = E[Y_i] = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_k x_{ik}$$

Multiple Linear Regression

$$\hat{y}_i = E[Y_i] = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_k x_{ik}$$

- If we let $B_0, B_1, \dots, B_k$ denote estimators of $\beta_0, \dots, \beta_k$, then the **sum of the squared differences between the Y_i and their estimated expected values $\hat{y}_i$** is

$$\sum_{i=1}^n (Y_i - B_0 - B_1 x_{i1} - B_2 x_{i2} - \cdots - B_k x_{ik})^2$$

- The least squares estimators are those values of $B_0, B_1, \dots, B_k$ that minimize the foregoing.

Multiple Linear Regression

- To determine the least squares estimators, we repeatedly take partial derivatives of the preceding sum of squares first with respect to B_0 , then to B_1 , . . . , then to B_k .
- On equating these $k + 1$ equations to 0, we obtain the following set of equations:

Multiple Linear Regression

$$\sum_{i=1}^n (Y_i - B_0 - B_1 x_{i1} - B_2 x_{i2} - \cdots - B_k x_{ik}) = 0$$

$$\sum_{i=1}^n x_{i1} (Y_i - B_0 - B_1 x_{i1} - \cdots - B_k x_{ik}) = 0$$

$$\sum_{i=1}^n x_{i2} (Y_i - B_0 - B_1 x_{i1} - \cdots - B_k x_{ik}) = 0$$

$\vdots$

$$\sum_{i=1}^n x_{ik} (Y_i - B_0 - B_1 x_{i1} - \cdots - B_k x_{ik}) = 0$$

Multiple Linear Regression

Rewriting these equations yields that the least squares estimators $B_0, B_1, \dots, B_k$ satisfy the following set of linear equations, called the *normal equations*:

$$\sum_{i=1}^n Y_i = nB_0 + B_1 \sum_{i=1}^n x_{i1} + B_2 \sum_{i=1}^n x_{i2} + \cdots + B_k \sum_{i=1}^n x_{ik} \quad (9.10.1)$$

$$\sum_{i=1}^n x_{i1} Y_i = B_0 \sum_{i=1}^n x_{i1} + B_1 \sum_{i=1}^n x_{i1}^2 + B_2 \sum_{i=1}^n x_{i1} x_{i2} + \cdots + B_k \sum_{i=1}^n x_{i1} x_{ik}$$

$\vdots$

$$\sum_{i=1}^n x_{ik} Y_i = B_0 \sum_{i=1}^n x_{ik} + B_1 \sum_{i=1}^n x_{ik} x_{i1} + B_2 \sum_{i=1}^n x_{ik} x_{i2} + \cdots + B_k \sum_{i=1}^n x_{ik}^2$$

Multiple Linear Regression

Before solving the normal equations, it is convenient to introduce matrix notation. If we let

$$\mathbf{Y} = \begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix}, \quad \mathbf{X} = \begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1k} \\ 1 & x_{21} & x_{22} & \cdots & x_{2k} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{nk} \end{bmatrix}, \quad \boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{bmatrix}, \quad \mathbf{e} = \begin{bmatrix} e_1 \\ e_2 \\ \vdots \\ e_n \end{bmatrix}$$

then $\mathbf{Y}$ is an $n \times 1$, $\mathbf{X}$ an $n \times p$, $\boldsymbol{\beta}$ a $p \times 1$, and $\mathbf{e}$ an $n \times 1$ matrix where $p \equiv k + 1$.

The multiple regression model can now be written as

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{e}$$

Multiple Linear Regression

In addition, if we let

$$\mathbf{B} = \begin{bmatrix} B_0 \\ B_1 \\ \vdots \\ B_k \end{bmatrix}$$

be the matrix of least squares estimators, then the normal Equations can be written as

$$\mathbf{X}'\mathbf{X}\mathbf{B} = \mathbf{X}'\mathbf{Y}$$

where $\mathbf{X}'$ is the transpose of $\mathbf{X}$. Assuming that $(\mathbf{X}'\mathbf{X})$ inverse exists, which is usually the case, we obtain, upon multiplying it by both sides of the foregoing, that the least squares estimators are given by

$$\mathbf{B} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$$

Multiple Linear Regression

Fit a multiple linear regression model to this data and predict the salary for an **employee with:**

Experience = 6 years

Education Level = PG (1)

Certifications = 3

Employee	Experience (Years)	Education Level (0 = UG, 1 = PG)	Certifications	Salary (₹1000s)
A	1	0	1	30
B	3	1	2	50
C	5	1	3	70
D	7	0	2	65
E	9	1	4	90

Multiple Linear Regression

$$Y = \beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k + e$$

$$\mathbf{B} = \begin{bmatrix} B_0 \\ B_1 \\ \vdots \\ B_k \end{bmatrix}$$

$$\mathbf{X}'\mathbf{X}\mathbf{B} = \mathbf{X}'\mathbf{Y}$$

$$\mathbf{B} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$$

Multiple Linear Regression

$$X'XB = X'Y$$

$$B = (X'X)^{-1}X'Y$$

$$X = \begin{bmatrix} 1 & 1 & 0 & 1 \\ 1 & 3 & 1 & 2 \\ 1 & 5 & 1 & 3 \\ 1 & 7 & 0 & 2 \\ 1 & 9 & 1 & 4 \end{bmatrix}$$

$$Y = \begin{bmatrix} 30 \\ 50 \\ 70 \\ 65 \\ 90 \end{bmatrix}$$

Employee	Experience (Years)	Education Level (0 = UG, 1 = PG)	Certifications	Salary (₹1000s)
A	1	0	1	30
B	3	1	2	50
C	5	1	3	70
D	7	0	2	65
E	9	1	4	90

Multiple Linear Regression

Day	Humidity (%)	Wind Speed (km/h)	Atmospheric Pressure (hPa)	Max Temperature (°C)
1	80	10	1015	28
2	75	12	1012	30
3	70	15	1010	32
4	65	18	1008	34
5	60	20	1005	36

Fit a multiple linear regression model and Predict the max temperature for a day with **65% humidity, 16 km/h wind speed, and 1009 hPa atmospheric pressure.**